Student Creation – User Guide

How a person becomes a student in OneCMS V2: menu order, one-time setup, the admission pipeline, and student login. Derived from a direct review of the frontend nav (`core/nav/nav-config.ts`), backend services and DB migrations.

> Menu items show only if the user holds at least one of the item's permission codes **and** the item's module is enabled for the deployment. Module gating always wins.

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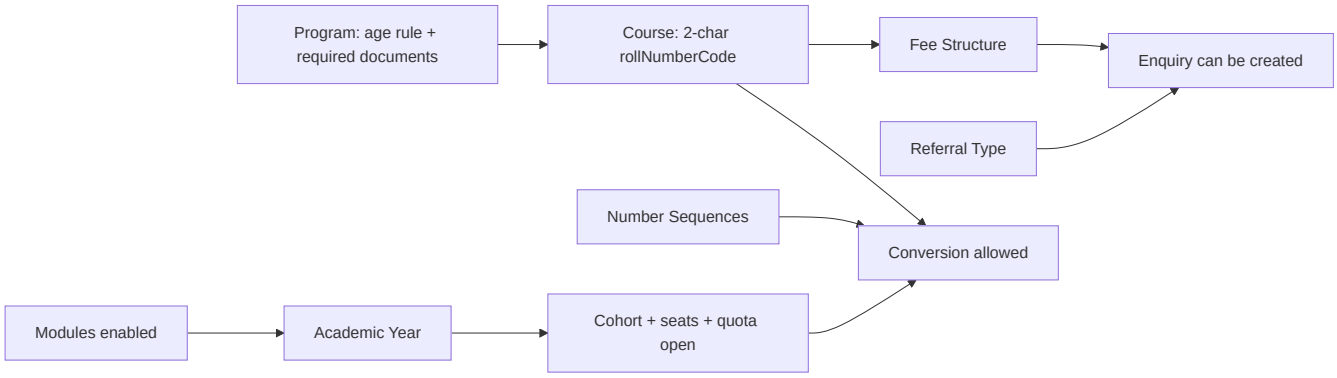
1. Sidebar menu order

#	Group (module)	Items in order
1	**Overview** (core)	Dashboard · My Profile · My Timetable (Student) · My Timetable (Staff) · My Dashboard · My Fees · My Wards · Ward Fees
2	**Admission Management** (ADMISSIONS)	Enquiries · Finalize Fee · Collect Payment · Submit Documents · Verify Documents · Complete Admission · Admission Explorer · AI Document Search · Retro Admit
3	**Student Management** (STUDENT_MGMT)	Student Explorer · AI Smart Search · Assign Roll Numbers · Scholarship Applications · Data Import
4	**Finance** (FINANCE)	Fee Explorer · Receipts · Refunds · Commissions
5	**Academics** (ACADEMICS)	Curriculum Versions · Syllabus · Experiments · CO/PO Mapping · Course Offerings · Elective Assignment · Capacity Auto-Plan · Assign Faculty · Recurring Unavailability · Timetable Builder · Timetable · Resource Timetable · Class Schedules · Faculty Absence · My Special Classes · Special Class Approvals · My Escort Duties · Attendance · Progress Report · Manage Exams · Exam Results · Student Promotion
6	**Library** (LIBRARY)	Issue Books · Issue Explorer · Overdue Books · Book Explorer · Journal Explorer · My Library · Fines · Racks & Shelves · Import · Library Settings
7	**Core Infrastructure** (CORE_INFRA)	Campus Infrastructure · Room Purpose Categories · Room Sub-Types
8	**Maintenance** (INVENTORY)	Maintenance Requests
9	**Stock Management** (INVENTORY)	Dashboard · Products · Categories · Units of Measure · Brands · UOM Conversion Templates · Locations · Storage Racks · Stock Balance · Cycle Counts · Stock Valuation · Goods Receipts · Stock Transfers · Supplier Returns · Reorder Configuration · Stock Indents · Loanable Item Issues
10	**Purchasing & Suppliers** (INVENTORY)	Suppliers · Rate Contracts · Vendor Product Rates · Price Comparison · Purchase Requisitions · Quotation Requests · Wanted List · Purchase Orders · PO Aging Report · PO Cycle-Time Report · Tax Rules · Currency Settings · Currency Exchange Rates
11	**Equipment & Asset Management** (INVENTORY)	Asset Register · Maintenance Schedules · Service Contracts · Depreciation Summary
12	**Budgets & Approvals** (INVENTORY)	Budgets · Budget vs. Actual · Approval Workflows · Approvals
13	**Gate Pass & Service Requests** (INVENTORY)	Gate Passes · Consignment Agreements · Consignment Stock · Service Tickets · Ticket Categories
14	**Hostel Management** (HOSTEL)	Hostel Room Types · Room Preferences · Room Allocation
15	**Reports & Analytics** (REPORTS)	General Reports · Fee Reports
16	**Preferences** (group ungated; items gated individually)	*Admissions:* Agents · Institutions · Referral Types · Staff Referrers. *Student:* Blood Groups · Communities · Scholarship Types. *Finance:* Fee Structures. *Academics:* Academic Calendar · Academic Years · Classrooms · Clinical Venues · Courses · Faculty · Faculty Doc Config · Faculty Workload Rules · Holiday Templates · Labs · Periods · Programs · Specialities · Subjects. *Inventory:* Equipment. *Core:* Designations · Location Master · Number Sequences · Settings
17	**User Management** (core)	Users · Roles & Permissions · Permission Tiers



2. One-time setup (before any admission)

#	What	Where	Notes
1	Enable modules	`app.modules.enabled` property	`ADMISSIONS`, `STUDENT_MGMT`, `FINANCE`, `ACADEMICS`. `FINANCE` requires `ACADEMICS`. Unset = all enabled.
2	Academic Year + Cohort	Preferences → Academic Years	Start year/date; initialize cohort for program/course/year.
3	Program	Preferences → Programs	Optional minimum age + cutoff day/month (enforced at enquiry and conversion). Set mandatory/optional document types.
4	Course	Preferences → Courses	**2-character `rollNumberCode` is required**; admission-number generation fails without it.
5	Cohort seats / quota	Academic Years	Total, management, counselling seats; quotas must be open. Conversion is blocked when closed or full (management overflow is allowed).
6	Fee Structure	Preferences → Fee Structures	Drives fee estimate and finalization. Finalization can only discount, never increase.
7	Referral Type	Preferences → Referral Types	Mandatory on an enquiry. Agents / Institutions / Staff Referrers optional.
8	Fee State, Location Master	Preferences	For enquiry fee-state and address fields.
9	Number Sequences	Preferences → Number Sequences	Admission, receipt, refund, commission series (seeded by migration).
10	Optional masters	Preferences	Blood Groups, Communities, Scholarship Types.

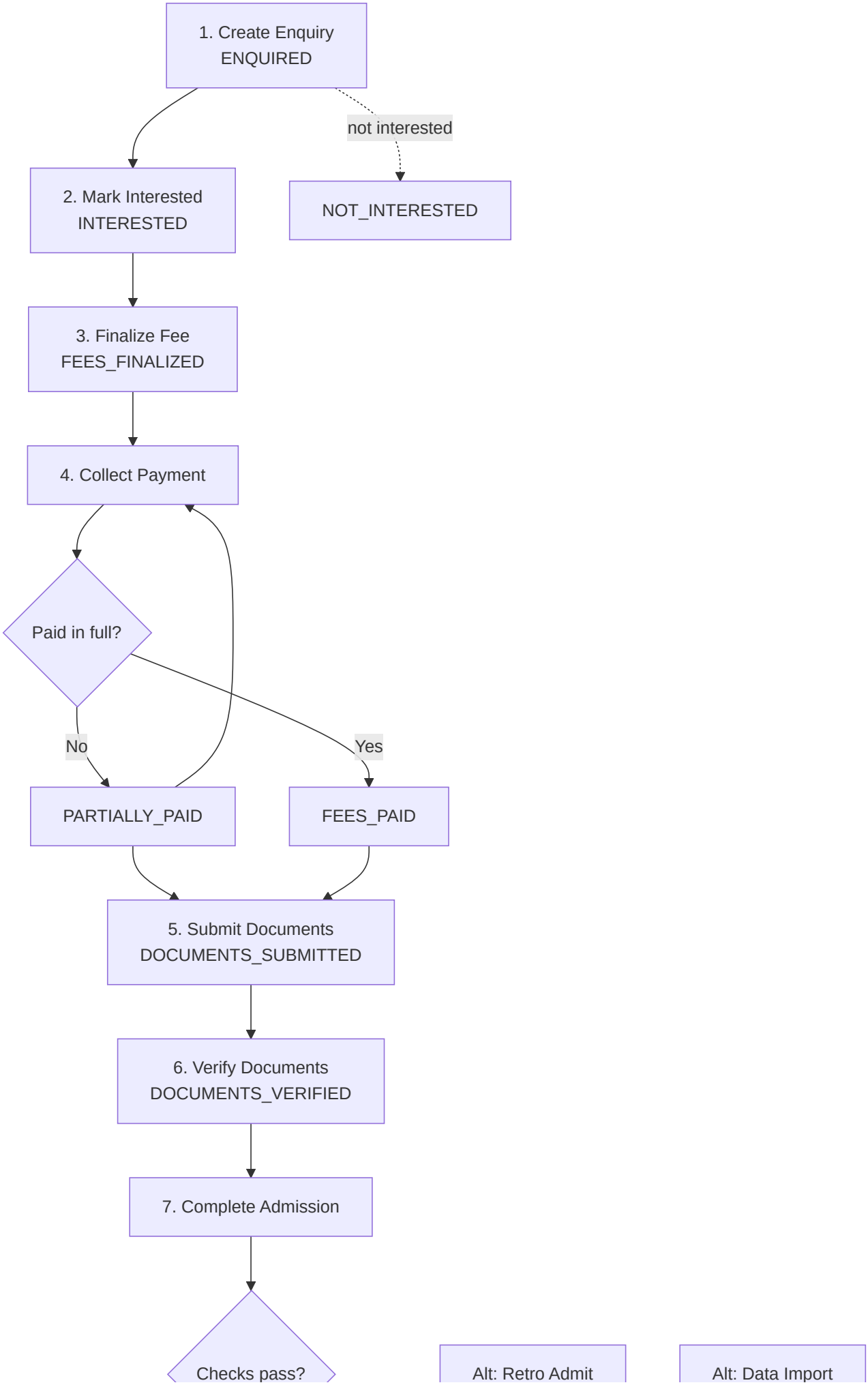


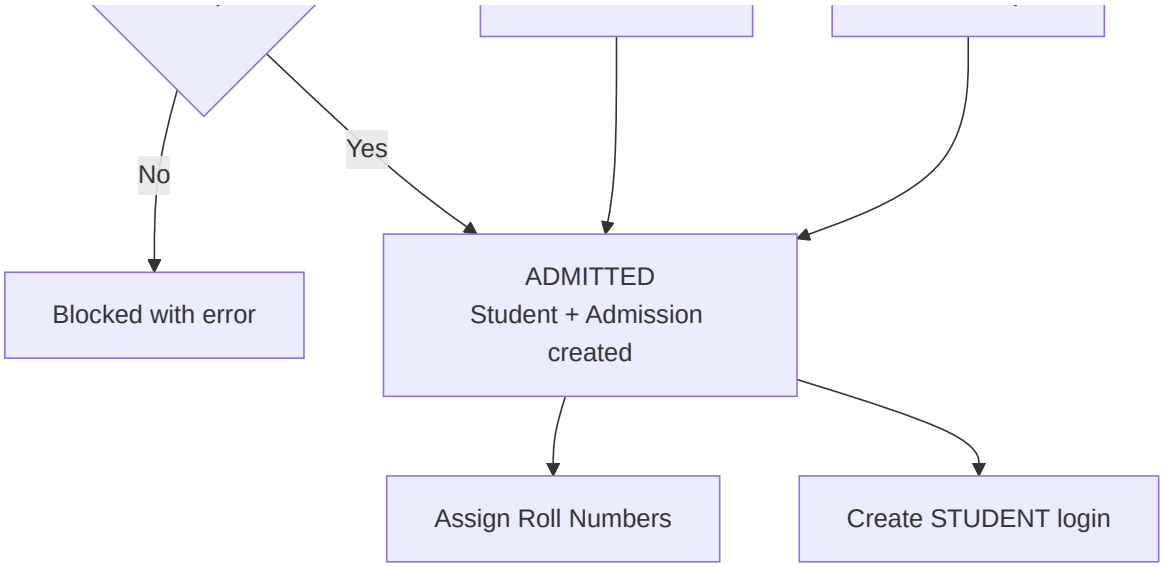
3. Admission pipeline

Step	Menu	Permission	Enquiry status after
1	Admission Management → Enquiries (create)	`ENQUIRY_CREATE`	ENQUIRED
2	Mark Interested (Program + Course required)	`ENQUIRY_EDIT`	INTERESTED
3	Finalize Fee (discount only; reason required)	`FEE_FINALIZE`	FEES_FINALIZED
4	Collect Payment	`FEE_COLLECT`	PARTIALLY_PAID → FEES_PAID when fully paid
5	Submit Documents (all mandatory present)	`DOCUMENT_SUBMISSION_MANAGE` + `ENQUIRY_EDIT`	DOCUMENTS_SUBMITTED

Step	Menu	Permission	Enquiry status after
6	Verify Documents	`DOCUMENT_VERIFICATION_MANAGE`	DOCUMENTS_VERIFIED (automatic once all mandatory docs verified)
7	Complete Admission (Convert form)	Menu: `ADMISSION_COMPLETE`; API: `ENQUIRY_EDIT`	**ADMITTED**

Step 7 validates: status is DOCUMENTS_VERIFIED, all mandatory documents verified, seat availability and quota open, unique student email, program age rule. It then creates the Student (ACTIVE) and Admission, generates the admission number (academic-year start year + course `rollNumberCode`), carries over the room preference, links enquiry documents, and auto-creates the fee allocation.


1. Create Enquiry
ENQUIRED
not interested
2. Mark Interested
INTERESTED
NOT_INTERESTED
3. Finalize Fee
FEES_FINALIZED
4. Collect Payment
Paid in full?
No
Yes
PARTIALLY_PAID
FEES_PAID
5. Submit Documents
DOCUMENTS_SUBMITTED
6. Verify Documents
DOCUMENTS_VERIFIED
7. Complete Admission
Checks pass?
Alt: Retro Admit
Alt: Data Import



Alternative routes

- **Retro Admit** (`RETRO_ADMIT`): for already-enrolled students. Needs unique email, roll number, university registration number, and at least one year of fees.
- **Data Import** (`IMPORT_DATA`): bulk import.

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4. After admission

- **Assign Roll Numbers** (Student Management): menu `ROLL_NUMBER_ASSIGN` , backend checks `STUDENT_CREATE` .
- **Cohort / section enrollment**: via Student Promotion, Elective Assignment and term enrollment.

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5. Student login

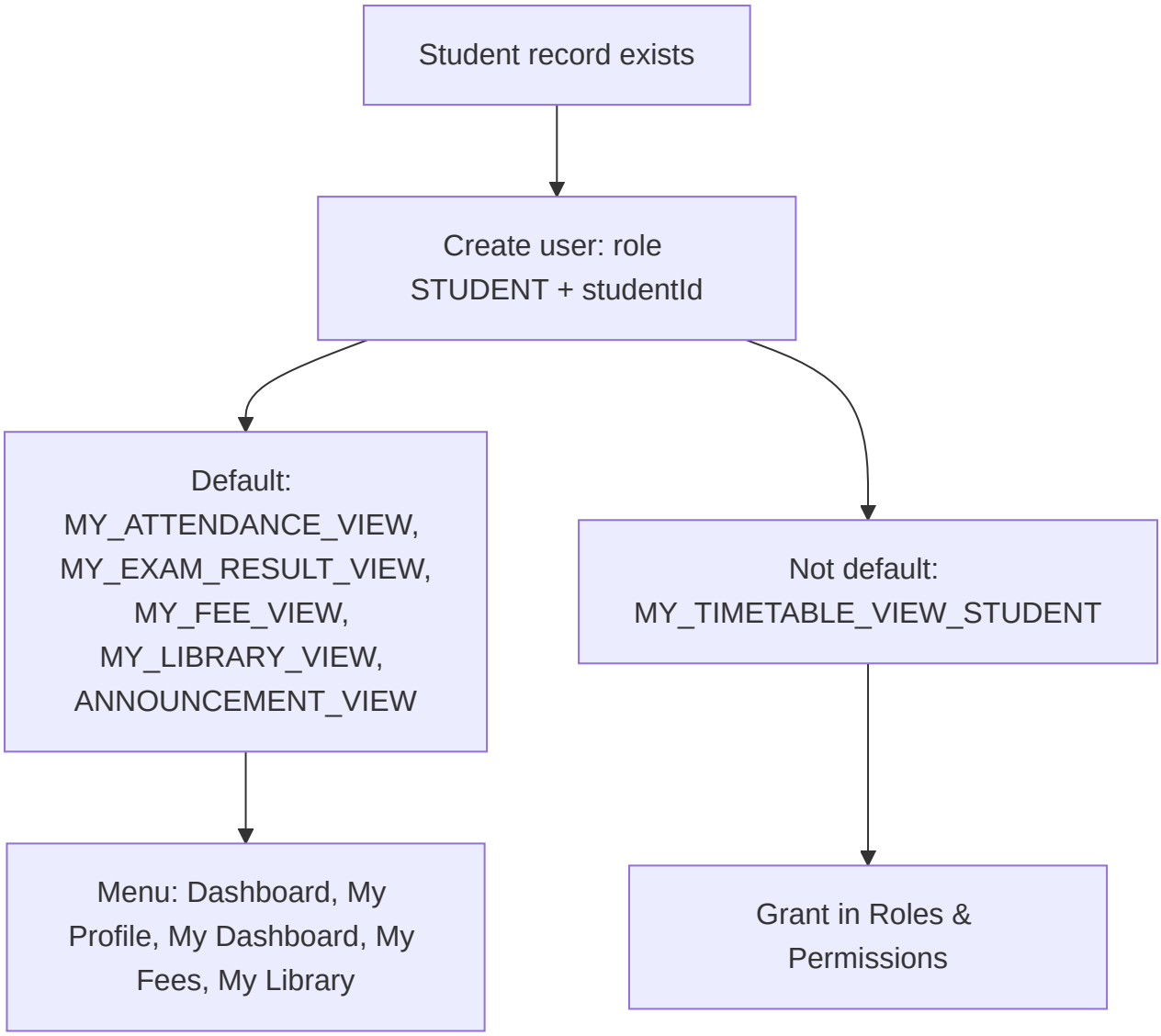
Admission does **not** create a login automatically.

1. User Management → Users → Create, role **STUDENT**.
2. The backend accepts `studentId` on user creation to link the account to the student record; all "My …" screens resolve the student through this link. The Users form does not expose a student picker, so the link is set via API (`POST /user-management` with `studentId` ; see `scripts/verify_student_portal_self_service.py`).
3. A Keycloak account is created with a temporary password; the student must change it at first login.

Default STUDENT role permissions:

Area	Permissions
Base	`STUDENT_VIEW` , `ATTENDANCE_VIEW` , `EXAM_RESULT_VIEW`
Library	`LIBRARY_CATALOGUE_VIEW` , `LIBRARY_ISSUE_VIEW` , `MY_LIBRARY_VIEW`
Self-service	`MY_ATTENDANCE_VIEW` , `MY_EXAM_RESULT_VIEW` , `MY_FEE_VIEW`
Announcements	`ANNOUNCEMENT_VIEW`

Resulting menu: Dashboard, My Profile, My Dashboard, My Fees, Library → My Library. **Not auto-granted:** `MY_TIMETABLE_VIEW_STUDENT` – grant it in Roles & Permissions to show "My Timetable (Student)".



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6. Known gotchas

- **Permission mismatch for Complete Admission:** menu `ADMISSION_COMPLETE`, route guard `ADMISSION_CREATE` / `ADMISSION_EDIT`, API `ENQUIRY_EDIT`. Give the admission officer role all of them.
- **STUDENT role may be missing:** migration `V125` deletes the seeded role when no users reference it. Verify under Roles & Permissions and create it if needed.
- **Student email must be unique**, or conversion fails.
- **Only ENQUIRED enquiries can be deleted.**